

# Zer0Miles

## Initial Database Design • V1

**Naming update:** In couples, the partner columns are **bear\_id** and **bunny\_id**, replacing user\_id\_a and user\_id\_b.

**Architecture:** Supabase Auth → PostgreSQL → RLS; selected tables use Realtime; photos use Supabase Storage.

Core relationship: auth.users → profiles → couples → daily\_questions/daily\_answers, moods, connection\_signals, love\_drops, daily\_photos, outfit\_colors. profiles also references avatars and pets.

## profiles

| Field        | Type        | Nullable | Default | Key | FK            |
|--------------|-------------|----------|---------|-----|---------------|
| id           | uuid        | No       | none    | PK  | auth.users.id |
| display_name | text        | No       | none    |     |               |
| avatar_id    | uuid        | Yes      | none    |     | avatars.id    |
| pet_id       | uuid        | Yes      | none    |     | pets.id       |
| created_at   | timestamptz | No       | now()   |     |               |
| updated_at   | timestamptz | No       | now()   |     |               |

• RLS ON • Realtime OFF • FK update Restrict • auth.users delete Cascade

## avatars

| Field      | Type        | Nullable | Default           | Key    | FK |
|------------|-------------|----------|-------------------|--------|----|
| id         | uuid        | No       | gen_random_uuid() | PK     |    |
| name       | text        | No       | none              |        |    |
| asset_key  | text        | No       | none              | UNIQUE |    |
| created_at | timestamptz | No       | now()             |        |    |

• RLS ON • Realtime OFF • Update/Delete Restrict

## pets

| Field      | Type        | Nullable | Default           | Key    | FK |
|------------|-------------|----------|-------------------|--------|----|
| id         | uuid        | No       | gen_random_uuid() | PK     |    |
| name       | text        | No       | none              |        |    |
| asset_key  | text        | No       | none              | UNIQUE |    |
| created_at | timestamptz | No       | now()             |        |    |

• RLS ON • Realtime OFF • Update/Delete Restrict

## couples

| Field      | Type        | Nullable | Default           | Key | FK          |
|------------|-------------|----------|-------------------|-----|-------------|
| id         | uuid        | No       | gen_random_uuid() | PK  |             |
| bear_id    | uuid        | No       | none              |     | profiles.id |
| bunny_id   | uuid        | No       | none              |     | profiles.id |
| created_at | timestamptz | No       | now()             |     |             |
| updated_at | timestamptz | No       | now()             |     |             |

• RLS ON • Realtime OFF • both FKs Update/Delete Restrict • CHECK bear\_id <> bunny\_id • no individual UNIQUE on bear\_id/bunny\_id

## daily\_questions

| Field         | Type       | Nullable | Default           | Key | FK          |
|---------------|------------|----------|-------------------|-----|-------------|
| id            | uuid       | No       | gen_random_uuid() | PK  |             |
| couple_id     | uuid       | No       | none              |     | couples.id  |
| question_date | date       | No       | none              |     |             |
| question_text | text       | No       | none              |     |             |
| created_by    | uuid       | No       | none              |     | profiles.id |
| created_at    | timestampz | No       | now()             |     |             |

• RLS ON • Realtime OFF • couple delete Cascade; created\_by delete Restrict • UNIQUE(couple\_id, question\_date)

## daily\_answers

| Field       | Type       | Nullable | Default           | Key | FK                 |
|-------------|------------|----------|-------------------|-----|--------------------|
| id          | uuid       | No       | gen_random_uuid() | PK  |                    |
| question_id | uuid       | No       | none              |     | daily_questions.id |
| user_id     | uuid       | No       | none              |     | profiles.id        |
| answer_text | text       | No       | none              |     |                    |
| answered_at | timestampz | No       | now()             |     |                    |

• RLS ON • Realtime OFF • question delete Cascade; user delete Restrict • UNIQUE(question\_id, user\_id) • answer hidden until both answer via later RLS/policies

## moods

| Field      | Type       | Nullable | Default           | Key | FK          |
|------------|------------|----------|-------------------|-----|-------------|
| id         | uuid       | No       | gen_random_uuid() | PK  |             |
| couple_id  | uuid       | No       | none              |     | couples.id  |
| user_id    | uuid       | No       | none              |     | profiles.id |
| mood       | text       | No       | none              |     |             |
| updated_at | timestampz | No       | now()             |     |             |

• RLS ON • Realtime ON • couple delete Cascade; user delete Restrict • UNIQUE(couple\_id, user\_id) • current mood only

## connection\_signals

| Field       | Type       | Nullable | Default           | Key | FK          |
|-------------|------------|----------|-------------------|-----|-------------|
| id          | uuid       | No       | gen_random_uuid() | PK  |             |
| couple_id   | uuid       | No       | none              |     | couples.id  |
| user_id     | uuid       | No       | none              |     | profiles.id |
| signal_type | text       | No       | none              |     |             |
| created_at  | timestampz | No       | now()             |     |             |
| expires_at  | timestampz | Yes      | none              |     |             |

• RLS ON • Realtime ON • couple delete Cascade; user delete Restrict • no UNIQUE/array

## love\_drops

| Field      | Type       | Nullable | Default           | Key | FK          |
|------------|------------|----------|-------------------|-----|-------------|
| id         | uuid       | No       | gen_random_uuid() | PK  |             |
| couple_id  | uuid       | No       | none              |     | couples.id  |
| sender_id  | uuid       | No       | none              |     | profiles.id |
| type       | text       | No       | none              |     |             |
| message    | text       | Yes      | none              |     |             |
| created_at | timestampz | No       | now()             |     |             |

• RLS ON • Realtime ON • couple delete Cascade; sender delete Restrict • no UNIQUE/array

## daily\_photos

| Field        | Type       | Nullable | Default           | Key | FK          |
|--------------|------------|----------|-------------------|-----|-------------|
| id           | uuid       | No       | gen_random_uuid() | PK  |             |
| couple_id    | uuid       | No       | none              |     |             |
| user_id      | uuid       | No       | none              |     | profiles.id |
| photo_date   | date       | No       | none              |     |             |
| storage_path | text       | No       | none              |     |             |
| created_at   | timestampz | No       | now()             |     |             |

• RLS ON • Realtime OFF • couple delete Cascade; user delete Restrict • actual image in Storage

## outfit\_colors

| Field      | Type       | Nullable | Default           | Key | FK          |
|------------|------------|----------|-------------------|-----|-------------|
| id         | uuid       | No       | gen_random_uuid() | PK  |             |
| couple_id  | uuid       | No       | none              |     | couples.id  |
| user_id    | uuid       | No       | none              |     | profiles.id |
| color      | text       | No       | none              |     |             |
| color_date | date       | No       | none              |     |             |
| created_at | timestampz | No       | now()             |     |             |

• RLS ON • Realtime OFF • couple delete Cascade; user delete Restrict • UNIQUE(couple\_id,user\_id,color\_date)

## RLS / Realtime Summary

| Field              | Type | Nullable | Default | Key | FK |
|--------------------|------|----------|---------|-----|----|
| profiles           | ON   | OFF      |         |     |    |
| avatars            | ON   | OFF      |         |     |    |
| pets               | ON   | OFF      |         |     |    |
| couples            | ON   | OFF      |         |     |    |
| daily_questions    | ON   | OFF      |         |     |    |
| daily_answers      | ON   | OFF      |         |     |    |
| moods              | ON   | ON       |         |     |    |
| connection_signals | ON   | ON       |         |     |    |
| love_drops         | ON   | ON       |         |     |    |
| daily_photos       | ON   | OFF      |         |     |    |
| outfit_colors      | ON   | OFF      |         |     |    |

All application tables use RLS. Realtime is limited to moods, connection\_signals, and love\_drops because these are the V1 features that benefit from immediate partner-side updates.

## Next database work

- Prevent a user from belonging to multiple active couples.
- Add/check indexes for common couple/user/date lookups.
- Implement RLS policies, including the daily-answer reveal rule.
- Configure Supabase Storage and photo access policies.
- Add updated\_at trigger behavior where required.
- Test unauthorized access using two separate users/couples.

## Design principles

PostgreSQL models relational data; UUIDs identify records; foreign keys enforce referential integrity; RLS is the client-facing authorization boundary; Realtime is selective; Storage holds image bytes while PostgreSQL holds metadata. V1 intentionally models a couple as exactly two people using bear\_id and bunny\_id.